St. George's University
Grenada, West Indies

Basic Principles of Medicine 1

Musculoskeletal System DLA 6

Thoracic Wall and Pectoral Region

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Recommended Reading

Drake, Vogl, Mitchell. Gray's Anatomy for students. 4th ed. Elsevier.

- Printed Textbook: Chapter 3 pg 140-168
- Online Textbook: Chapter 3
  - Conceptual Overview
  - Regional Anatomy
    - Pectoral region
    - Thoracic wall
    - Diaphragm
    - Movements of the thoracic wall and diaphragm during breathing
    - Pleural cavities

Objectives

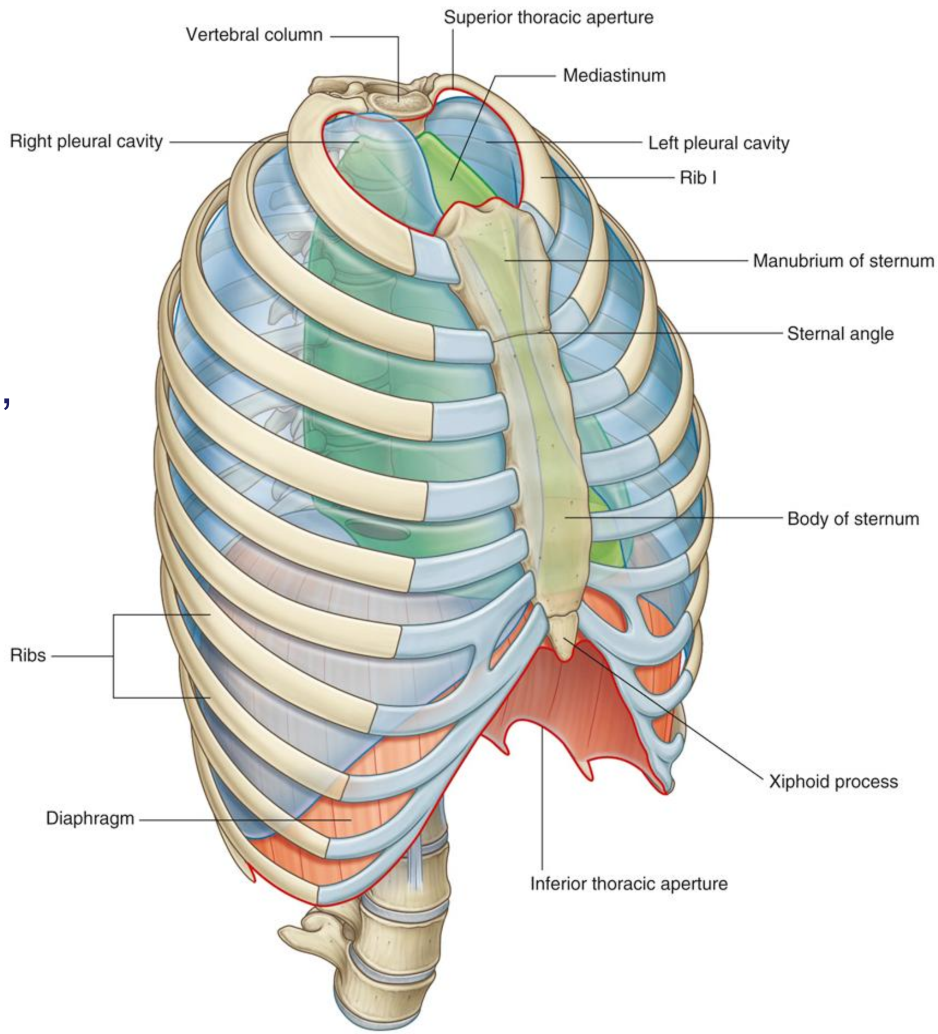
SOM.MKII.BPM1.1.MSK.1.ANAT.1212	Describe the skeletal framework of the thoracic wall.
SOM.MKII.BPM1.1.MSK.1.ANAT.1213	Describe the costovertebral, sternocostal and sternoclavicular joints.
SOM.MKII.BPM1.1.MSK.1.ANAT.1214	List the general attachments, innervations and actions of the intercostal muscles.
SOM.MKII.BPM1.1.MSK.1.ANAT.1215	Describe the course of the intercostal neurovascular bundle
SOM.MKII.BPM1.1.MSK.1.ANAT.1216	List the innervations, blood supply and actions of the muscles of the anterior and lateral chest wall
SOM.MKII.BPM1.1.MSK.1.ANAT.1217	Discuss and describe the bucket-handle and pump-handle movements of the ribs during respiration.
SOM.MKII.BPM1.1.MSK.1.ANAT.1218	Describe the dermatomes of the thoracic wall.

Objectives

SOM.MKII.BPM1.1.MSK.1.ANAT.1219	Describe the boundaries of the superior and inferior thoracic apertures.
SOM.MKII.BPM1.1.MSK.1.ANAT.1220	List the layers of the thoracic wall from skin to the parietal pleura.
SOM.MKII.BPM1.1.MSK.1.ANAT.1221	Describe the locations of the parietal pleura and its reflection lines.
SOM.MKII.BPM1.1.MSK.1.ANAT.1222	List the recesses formed by the pleural reflections.
SOM.MKII.BPM1.1.MSK.1.ANAT.1223	Describe the arterial supply, venous drainage and innervation of the parietal pleura.
SOM.MKII.BPM1.1.MSK.1.ANAT.1224	Describe the arterial supply, venous and lymphatic drainage of the thoracic wall.
SOM.MKII.BPM1.1.MSK.1.ANAT.1225	Describe the location, structure, innervation, blood supply and lymphatic drainage of the mammary gland.

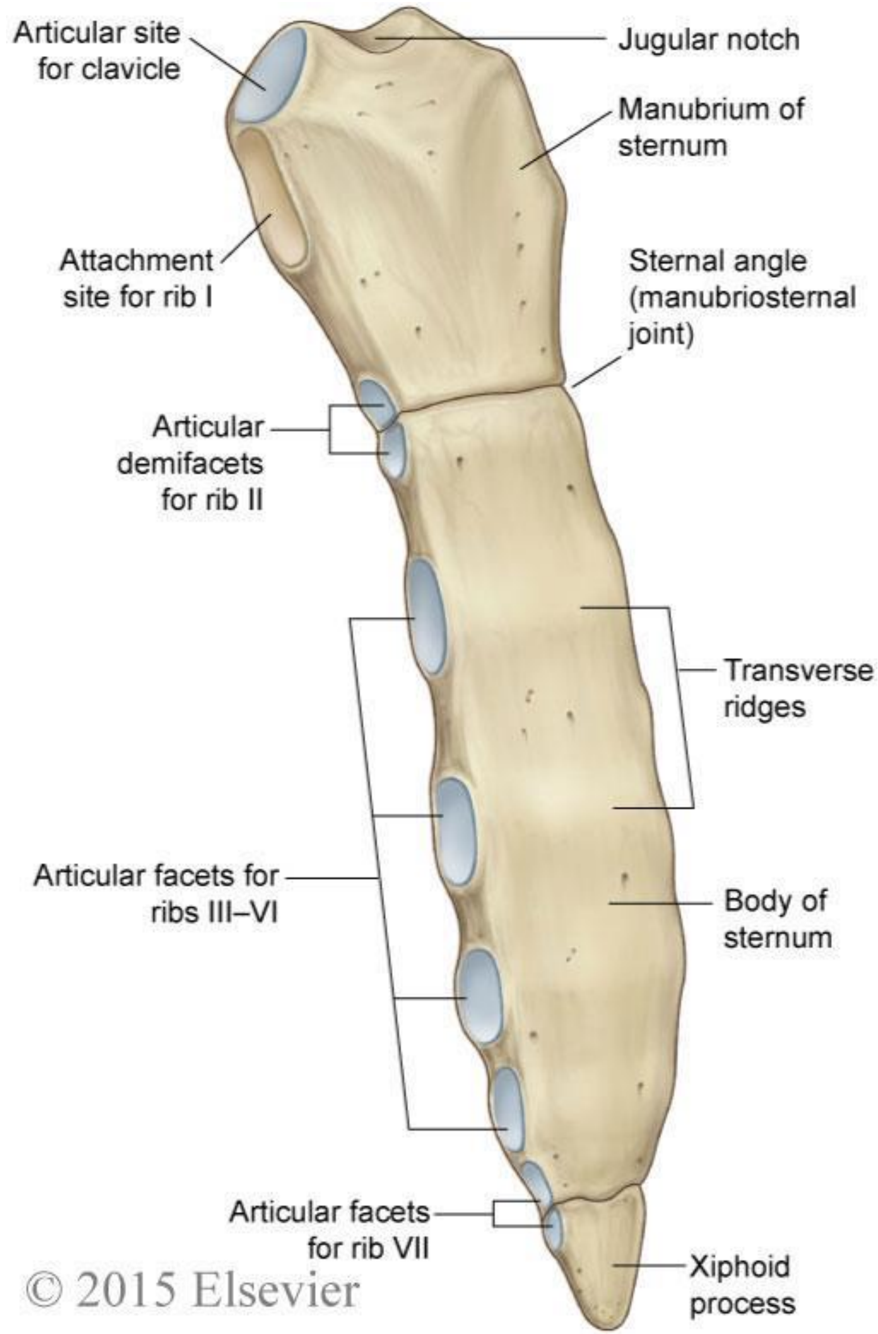
Thoracic wall

- Bones
  - Vertebrae, ribs, sternum
- Muscles
  - Diaphragm, intercostal, pectoral, serratus anterior, latissimus dorsi
- Nerves and blood vessels
  - Intercostal neurovascular bundle
- Other
  - Skin, subcutaneous tissue, endothoracic fascia, parietal pleura



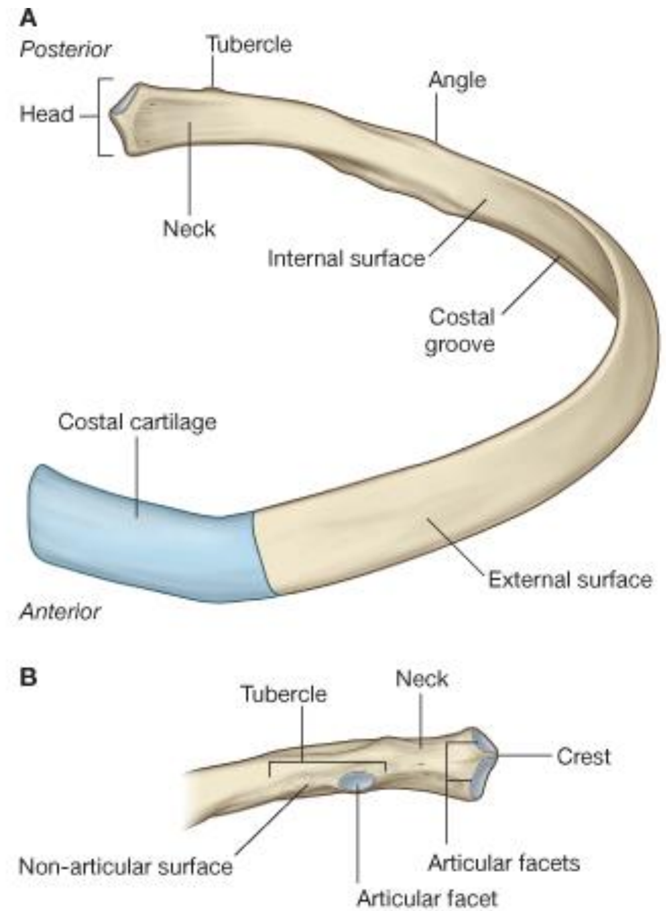
Sternum

- Consists of three portions:
  - Manubrium
  - Body
  - Xiphoid process
- Articulates with:
  - Clavicle
  - Costal cartilages I-VII
- **Jugular notch**
  - Small indentation on the superior border of the manubrium
- **Manubriosternal joint** (angle of Louis)
  - Anterior boundary of the transthoracic plane
  - Corresponds to:
    - T4/5 intervertebral disc
    - 2nd costal cartilage



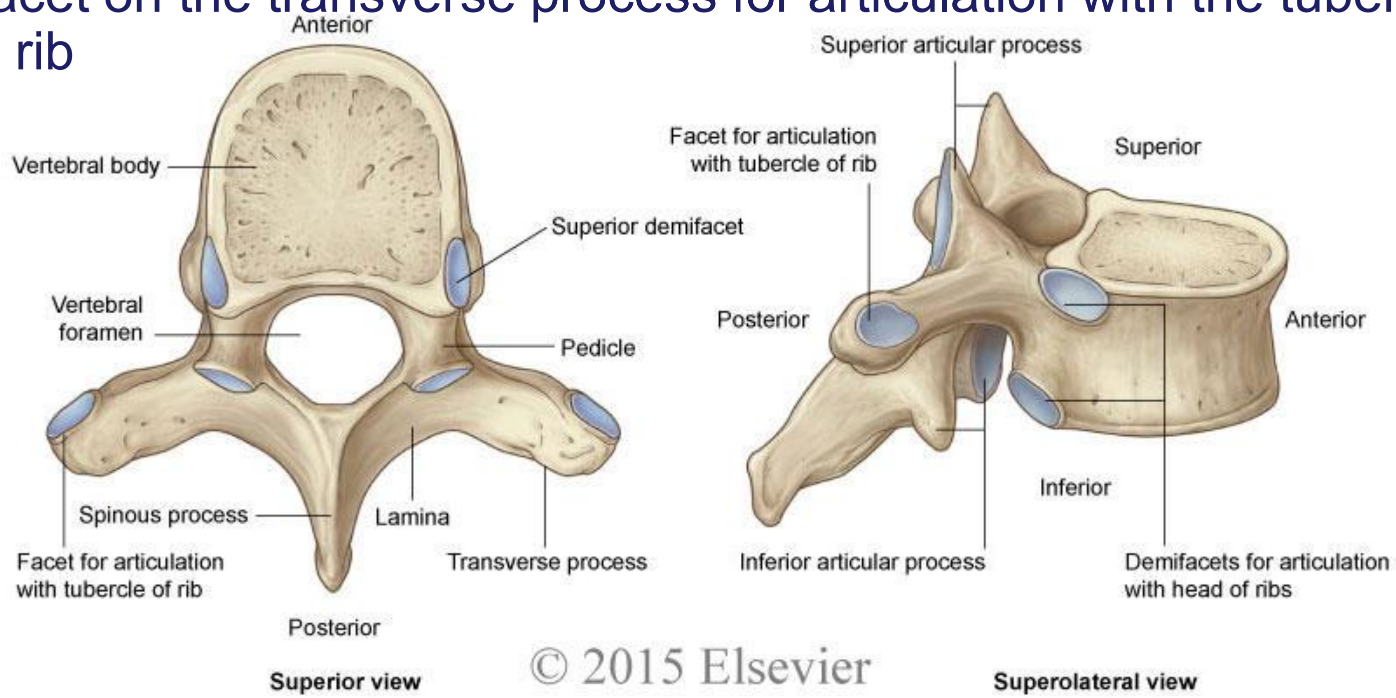
Ribs

- Typical ribs
  - Head
    - 2 articular facets
  - Tubercle
    - 1 articular facet
  - Angle
  - Costal groove – for neurovascular bundle
  - Sternal end – attached to costal cartilage
- Atypical ribs (1, 2, 10, 11, 12)
  - Each atypical for their own reason
  - Also correspond to the atypical vertebrae



Thoracic vertebrae

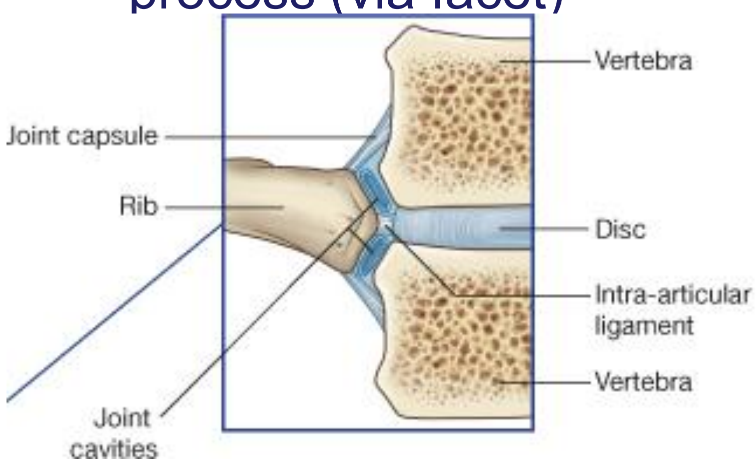
- Small heart shaped body
- Round vertebral foramen
- Downward slanting spinous process
- Two demifacets on body for articulation with the head of the rib
- A facet on the transverse process for articulation with the tubercle of the rib



Joints

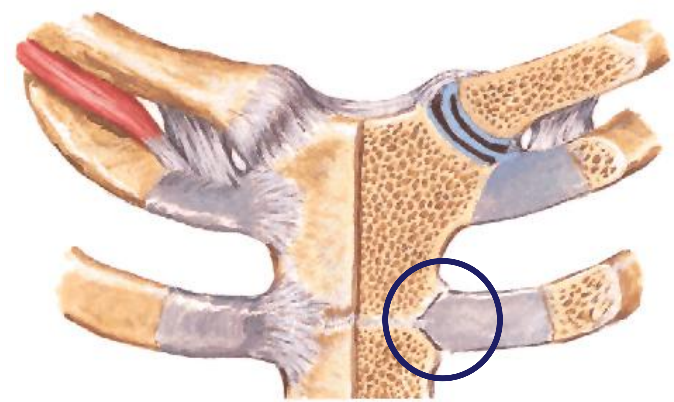
COSTOVERTEBRAL

- Synovial joints between rib and vertebrae
- Typical rib articulates with:
  - Adjacent vertebral body and body of vertebra above including disk (via demi facets)
  - Adjacent transverse process (via facet)



STERNOCOSTAL

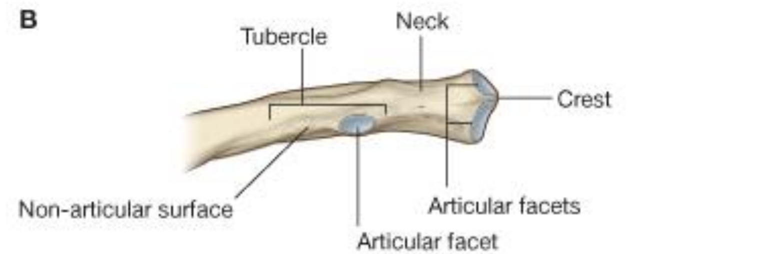
- Synovial joints between costal cartilage and sternum (1st is fibrocartilaginous)
- True ribs attach with sternum directly (via costal cartilage)
- False ribs attach indirectly by joining costal cartilage of rib 7
- Floating ribs don't attach to sternum

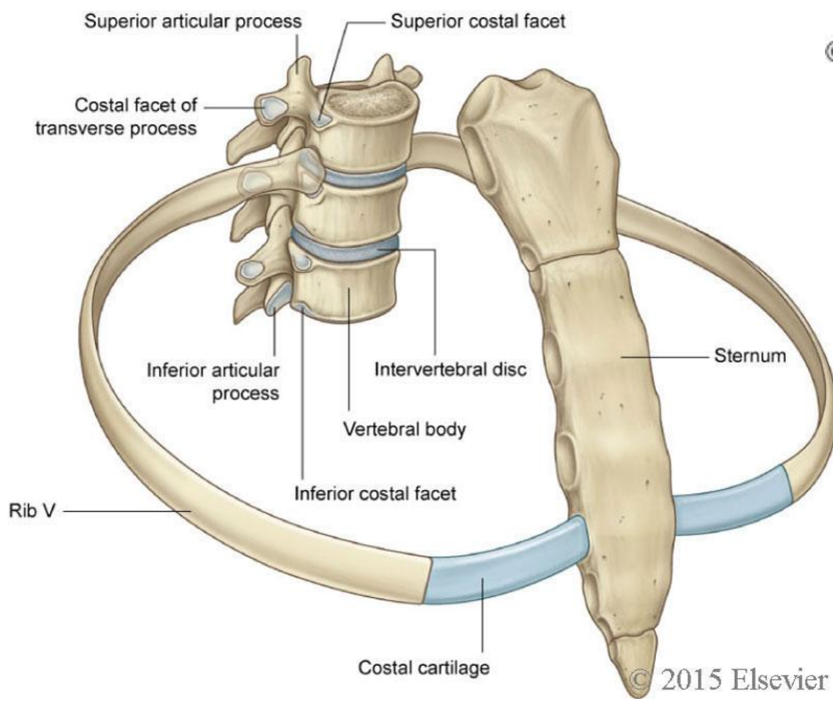


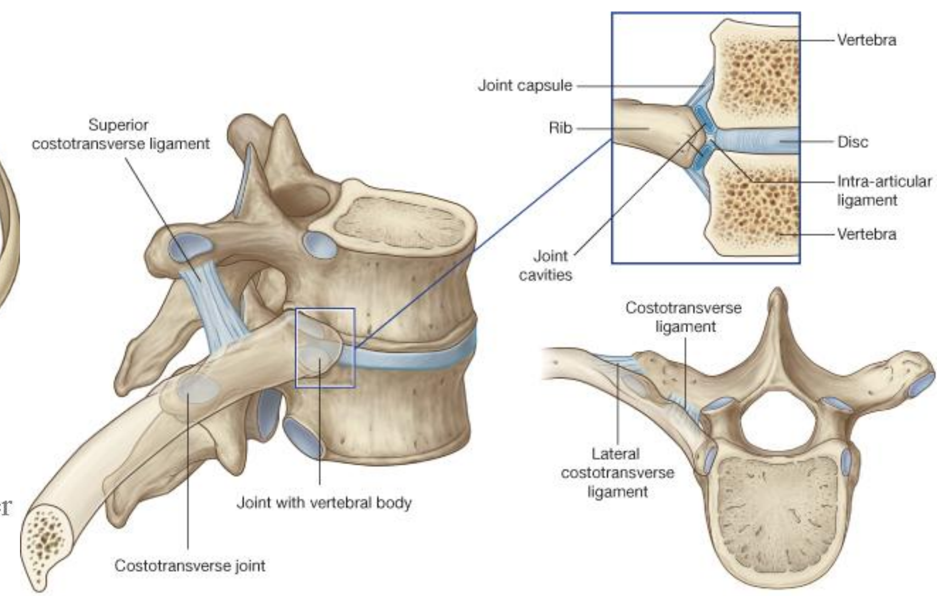
Netter Plate 419

Costovertebral and Sternocostal joints

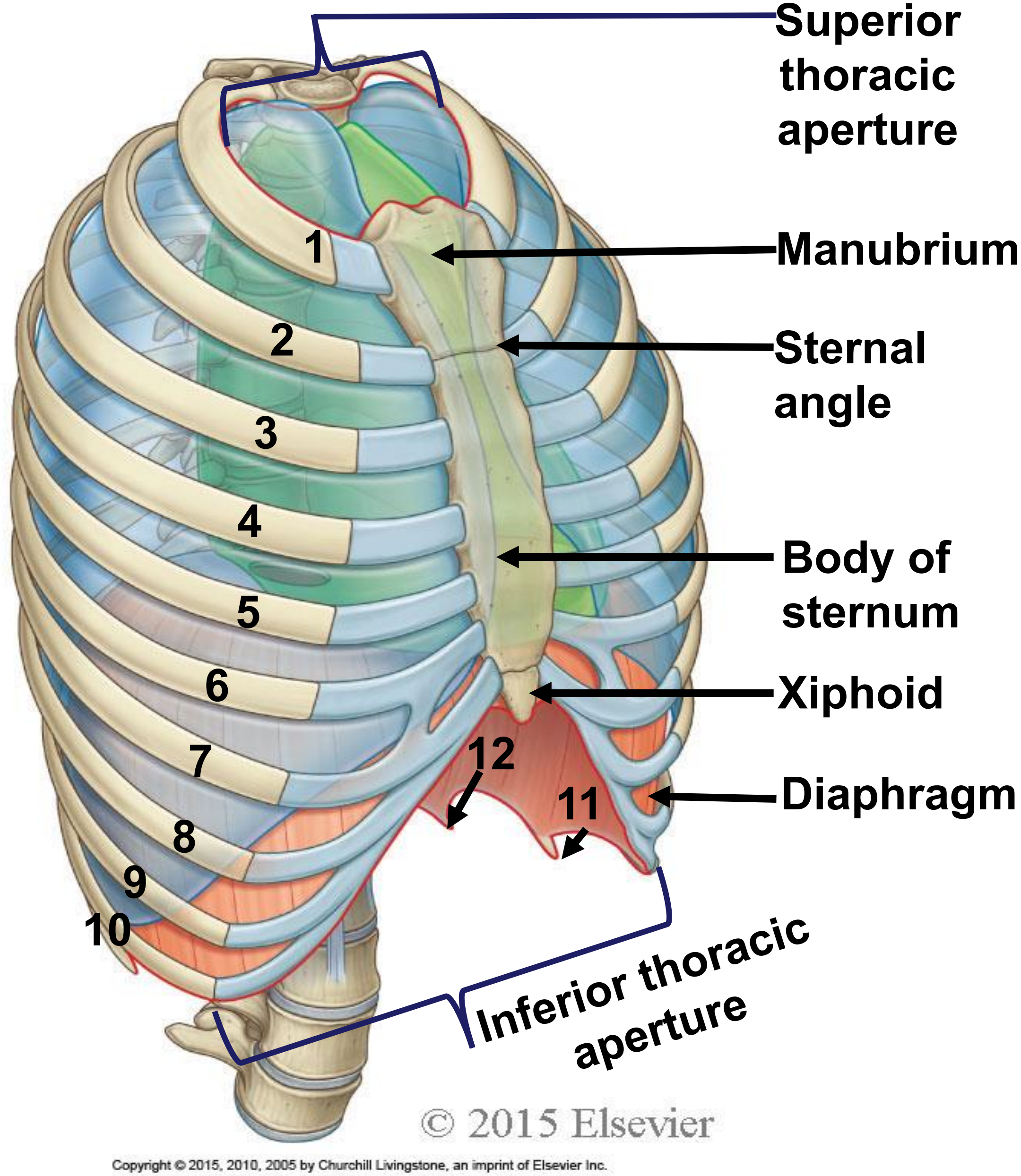
True ribs (1-7) - attach to sternum via own costal cartilage
False ribs (8-10) - attach to costal cartilage of rib above
Floating ribs - no attachment anteriorly





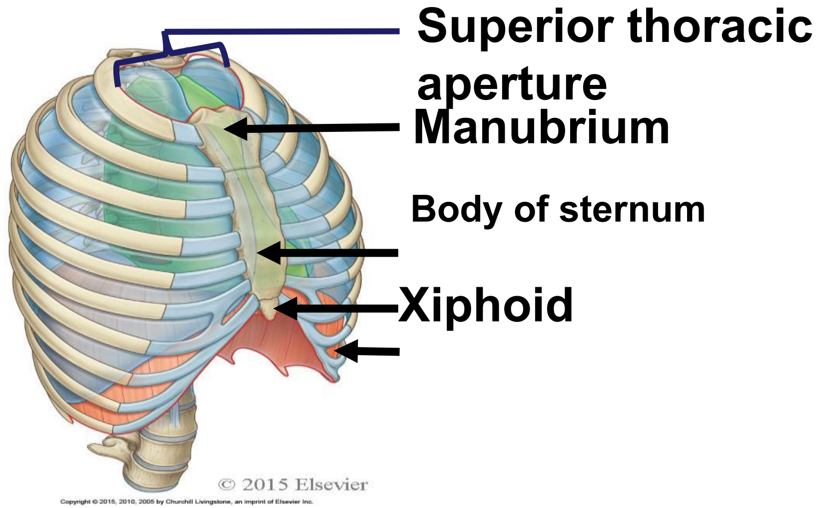


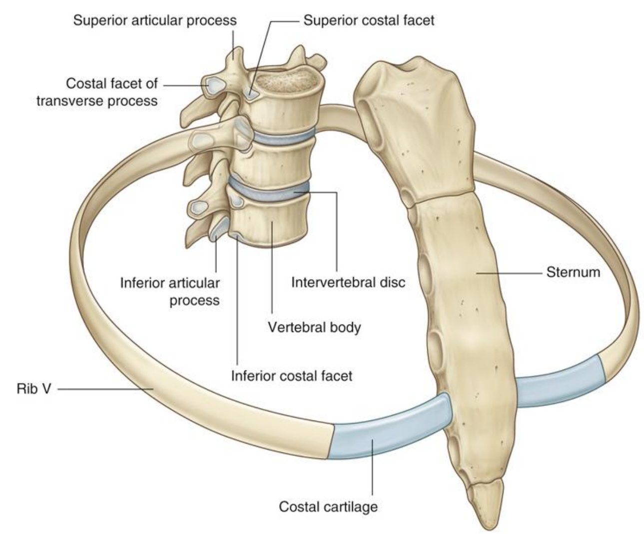
Skeletal Framework of the thoracic wall



Superior Thoracic Aperture-Boundaries

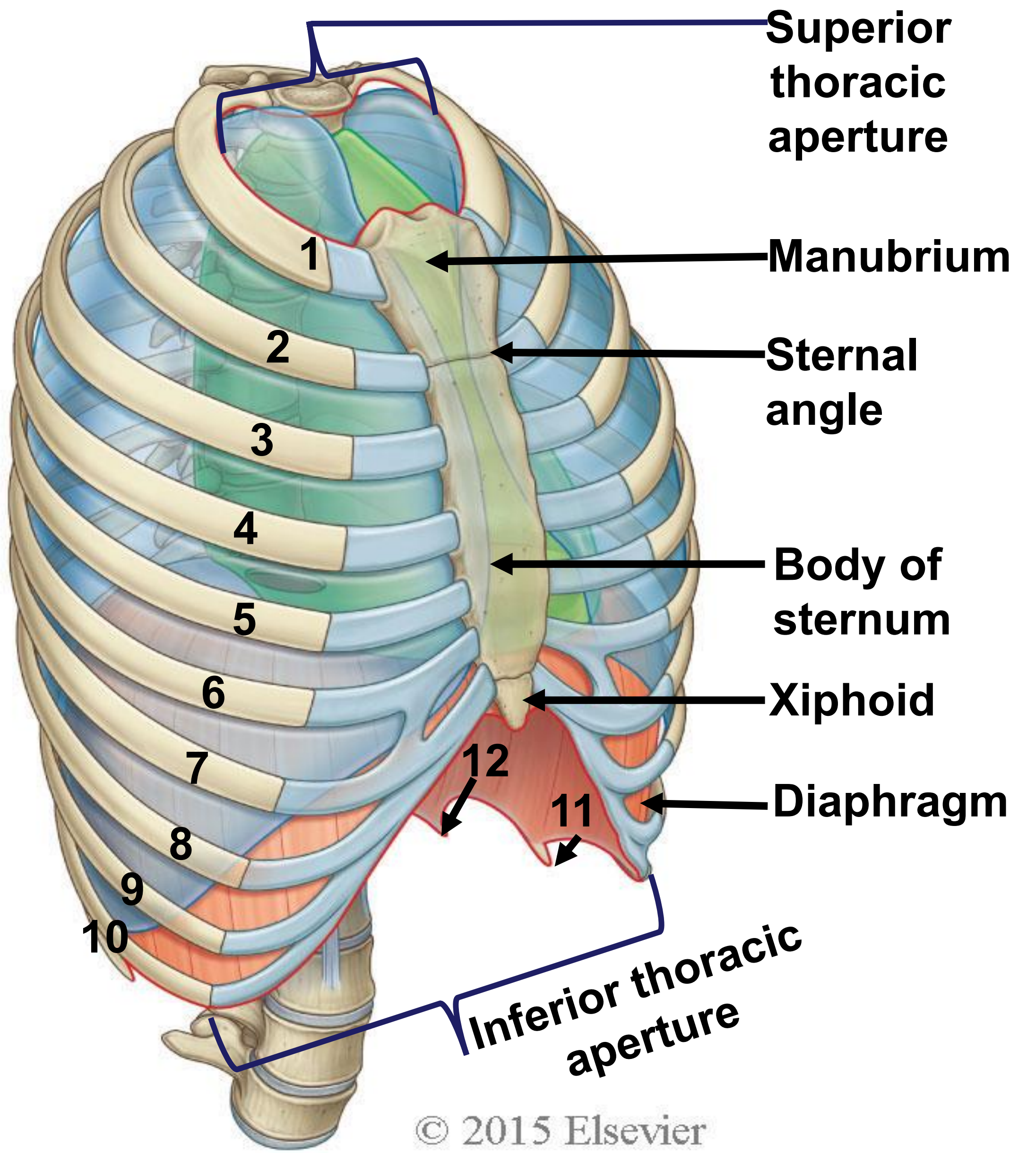
- Boundaries:
    - Body of 1st thoracic vertebra
    - 1st ribs and their costal cartilage
    - Jugular notch
- Structures passing through
    - Left Subclavian artery
    - Subclavian (or SCV) vein
    - Brachiocephalic trunk





Inferior Thoracic Aperture-Boundaries

- 12th thoracic vertebra
- 11th and 12th pair of ribs
- Costal cartilages of ribs
  7th – 10th (costal margin)
- Xiphisternal joint
- Closed by the diaphragm



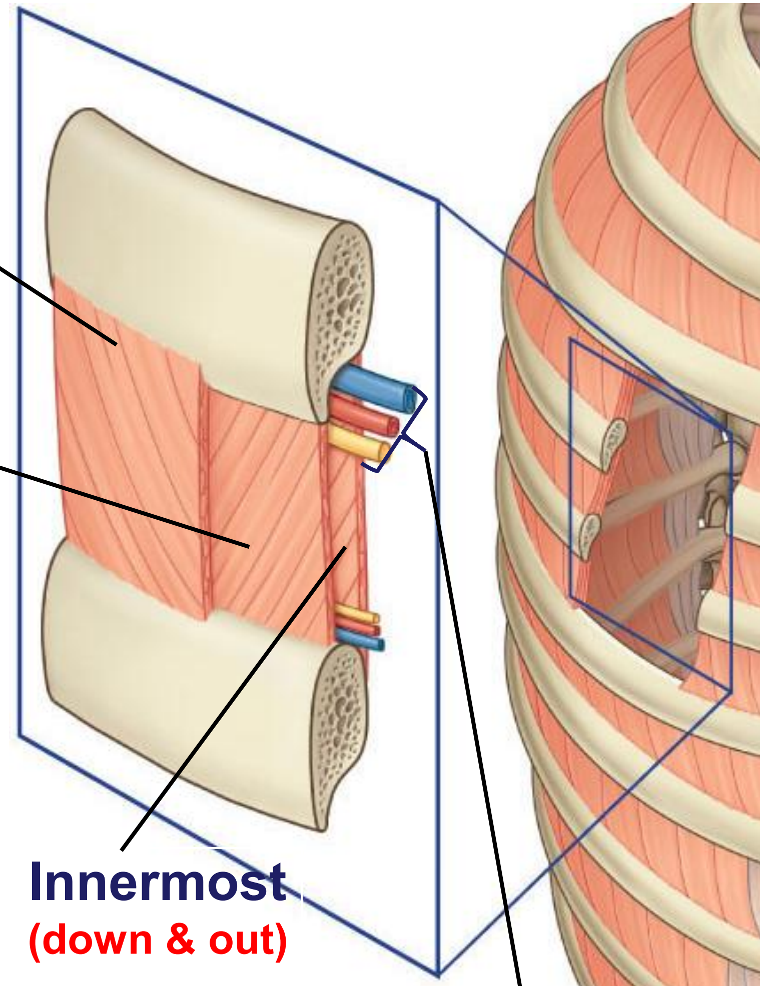
Thoracic and Pectoral Muscles

Intercostal muscles

- **External intercostal**
  - Fibers – down and medial
    “hands in the front pockets”
- **Internal intercostal**
  - Fibers – down and lateral
    “hands in the back pockets”
- **Innermost intercostal**
  - Fibers – same as internal
- Neurovascular bundle
  - intercostal artery, vein, nerve
  - between internal and innermost
- Action: Elevation and depression of ribs and

maintaining the intercostal space during respiration

Innervation is segmental from intercostal nerves*

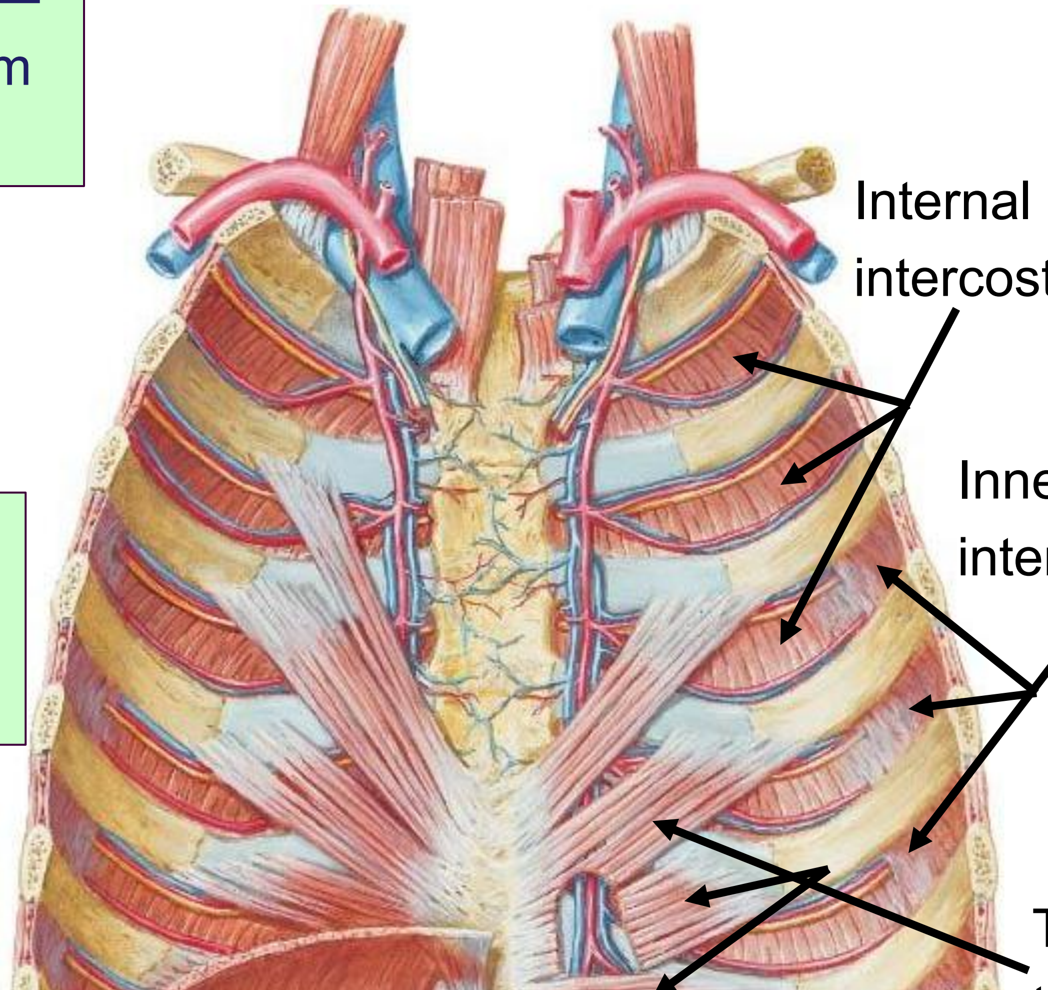


Deep muscles of the thoracic cage

Transversus thoracis muscle

"Radiates out from the sternum"

*****Innervation is segmental from intercostal nerves*****



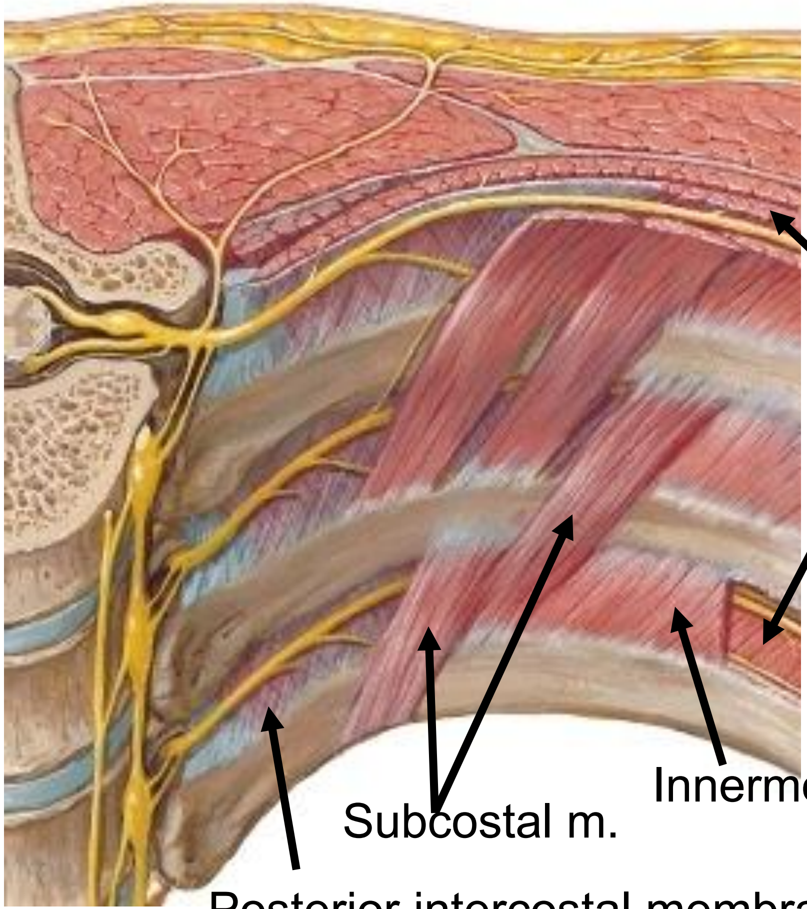
Anterior chest wall – posterior view

Deep muscles of the thoracic cage cont.

Subcostal muscles

Found posteriorly near the angles of the ribs and span 1 or more intercostal spaces

Posterior chest wall – anterior view

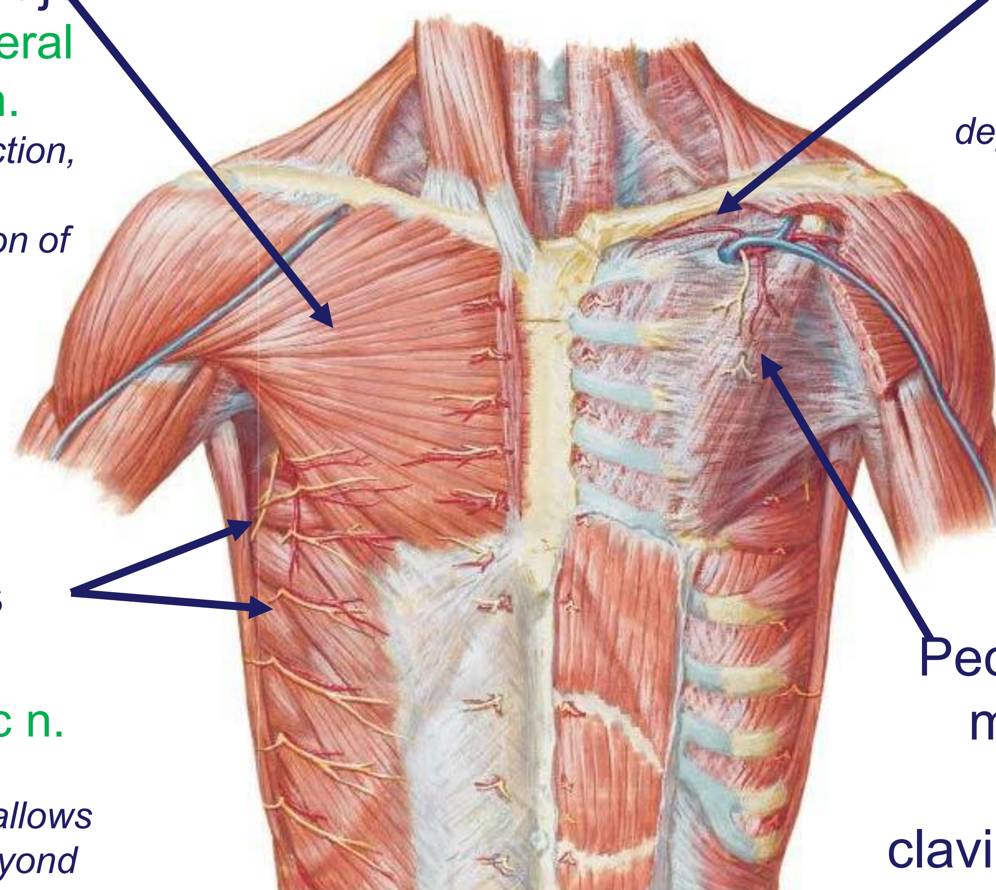


Innervation is segmental from intercostal nerves

Anterolateral thoracic wall muscles

Pectoralis major
medial & lateral pectoral n.
*Shoulder a**dd**uction, flexion & medial rotation of humerus*

Subclavius
nerve to subclavius
depresses clavicle



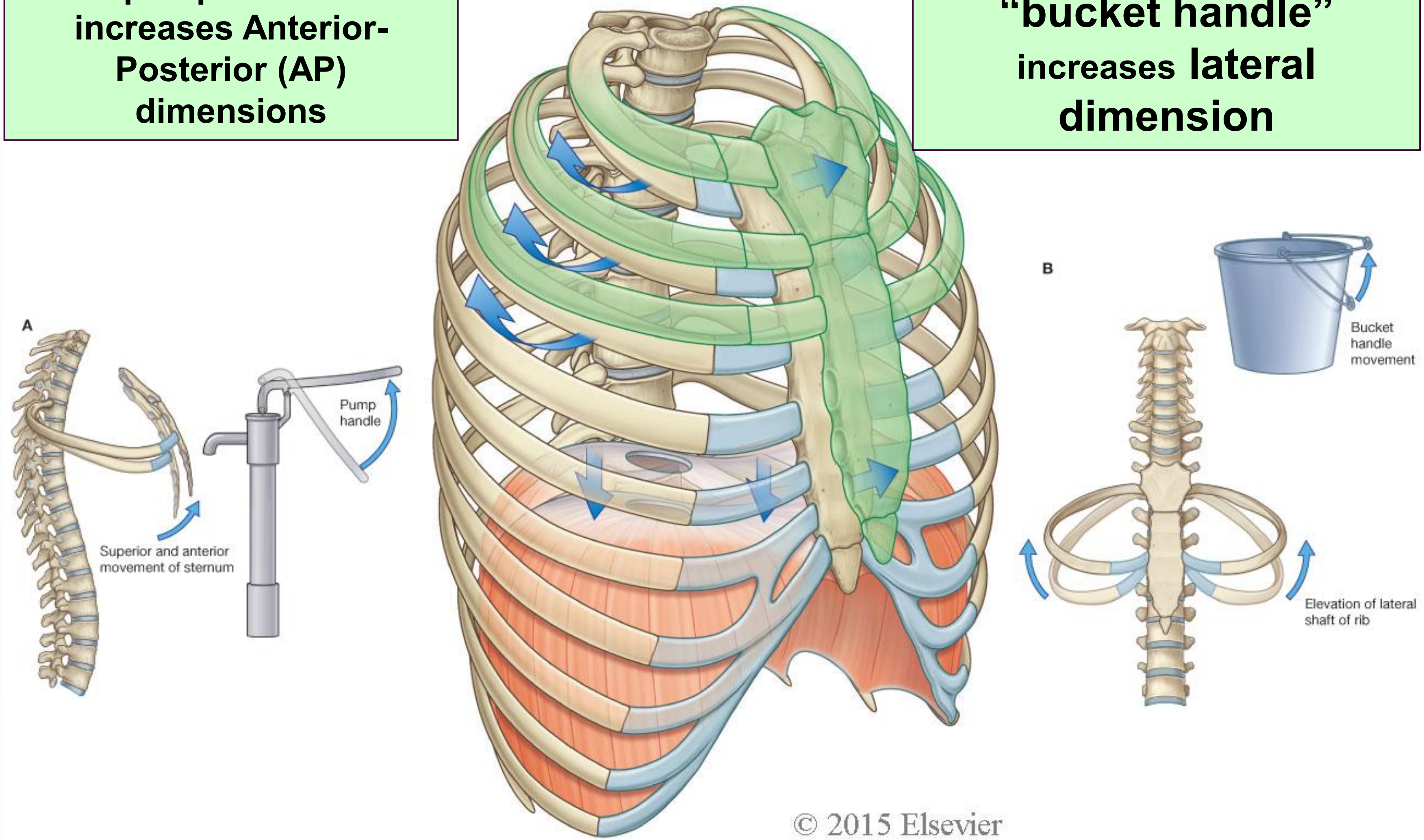
Serratus anterior
long thoracic n.
upward rotates scapula (allows for abduction beyond 90°)
Keeps scapula against thoracic cage

Pectoralis minor (in clavipectoral fascia)
medial pectoral n.
depresses shoulder, protracts scapula

Rib movement during respiration

"pump handle" increases Anterior-Posterior (AP) dimensions

"bucket handle" increases lateral dimension

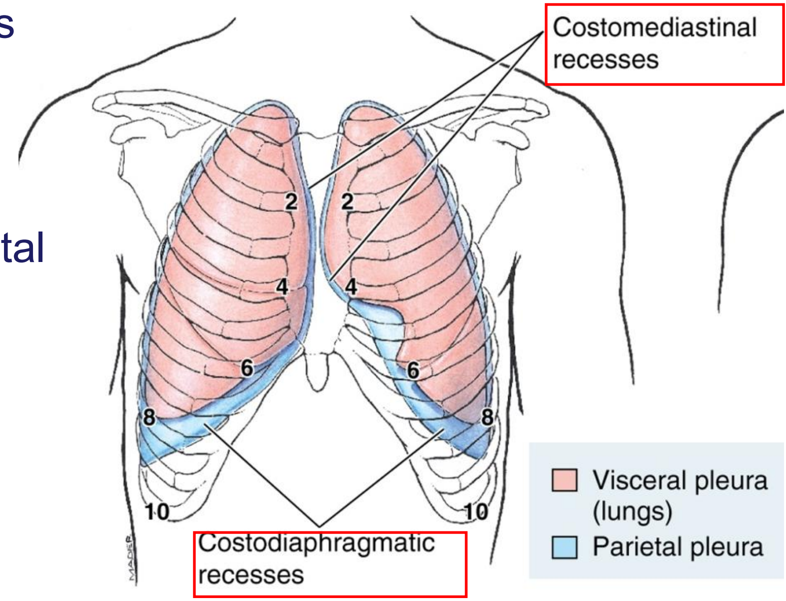


Contraction of diaphragm increases vertical dimension

Pleura

Pleura

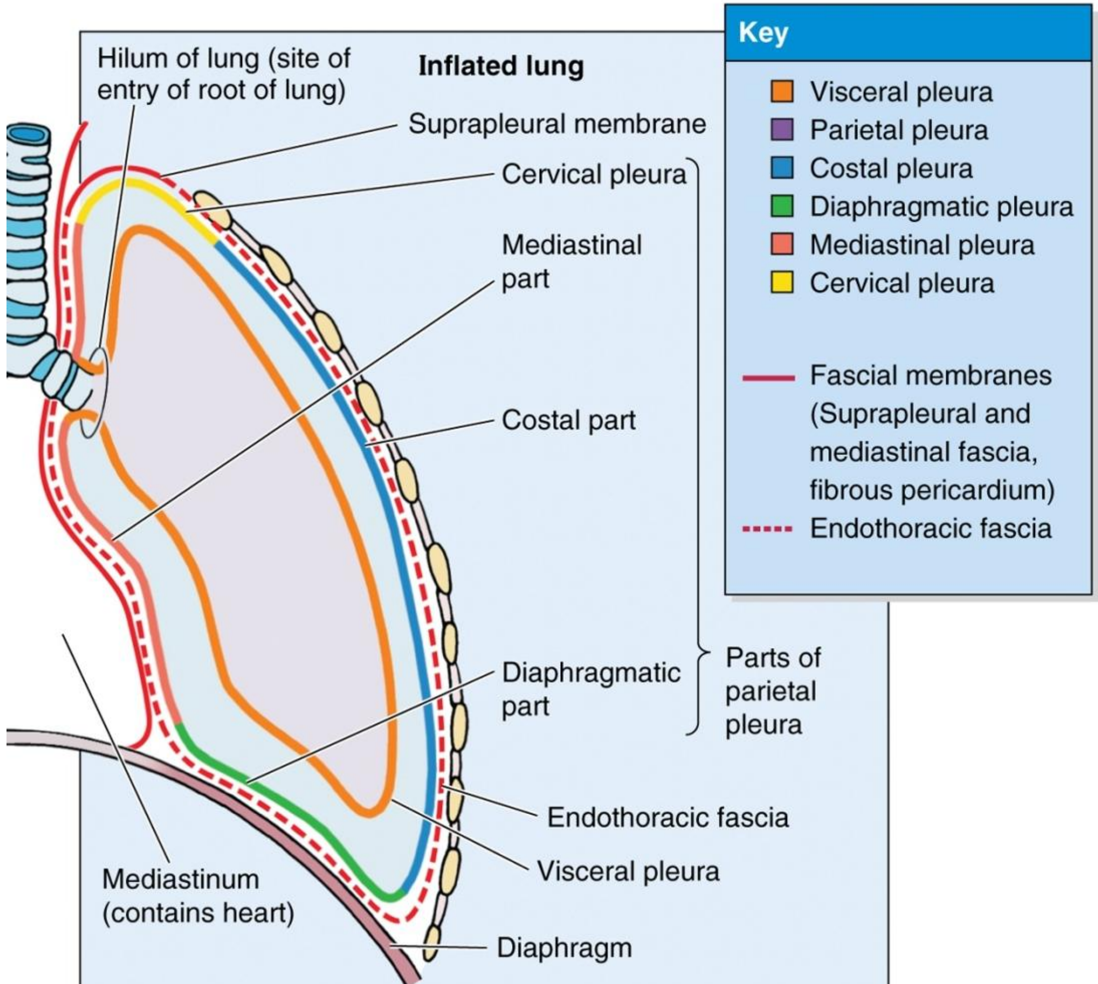
- Pleura are membranes consisting of two parts separated by a space (pleural cavity)
  - Visceral – lines/adhered to the lungs
  - Parietal – lines/adhered to the thoracic cage
    - Named according to location and receives innervation accordingly
      - Costal; intercostal nerves
      - Mediastinal; phrenic nerve
      - Diaphragmatic; phrenic and intercostal
      - Cervical;
    - Folds to conform to the boundaries of the thoracic cavity reflection lines are formed
    - Contains two **recesses** to allow for movement of the lungs during breathing
      - **Costodiaphragmatic**
      - **Costomediastinal**



- The lungs don't fill the entire space to allow for movement during breathing

The costodiaphragmatic recess is ideal location for thoracocentesis

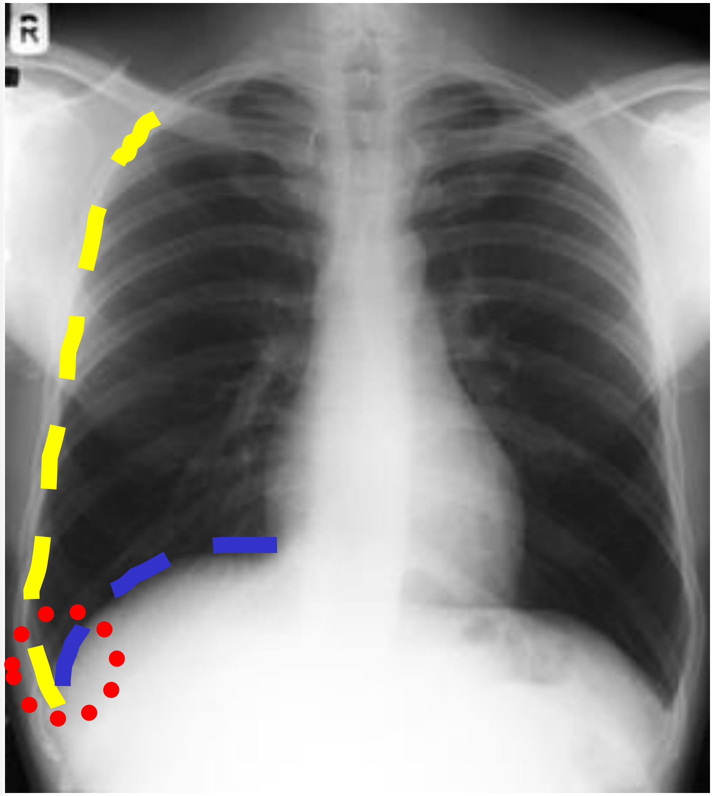
Parietal and visceral pleura



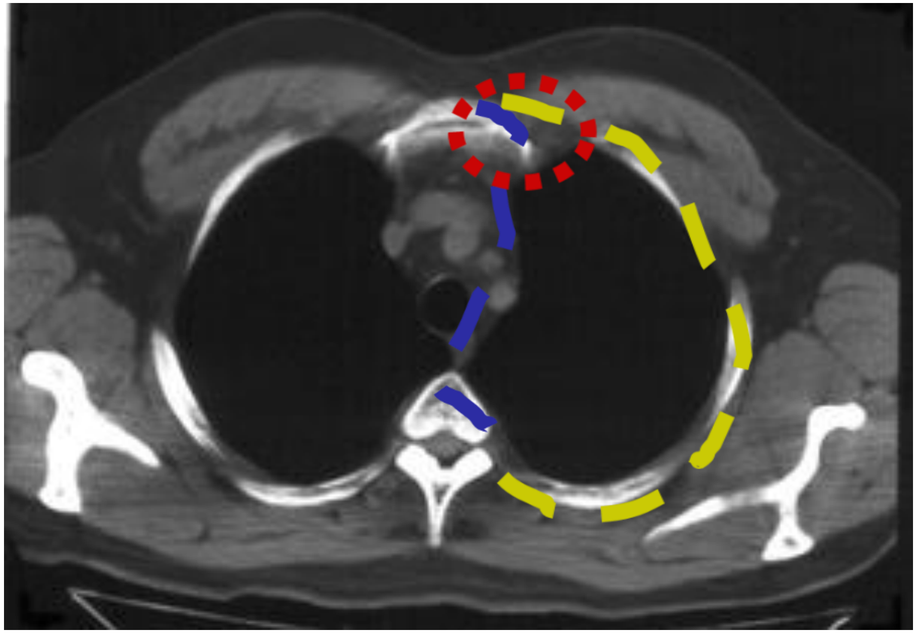
The right and left pulmonary cavities, separated by the mediastinum, are lined with parietal pleura that reflects onto the lungs as visceral pleura.

Pleural reflections (recesses)

COSTODIAPHRAGMATIC

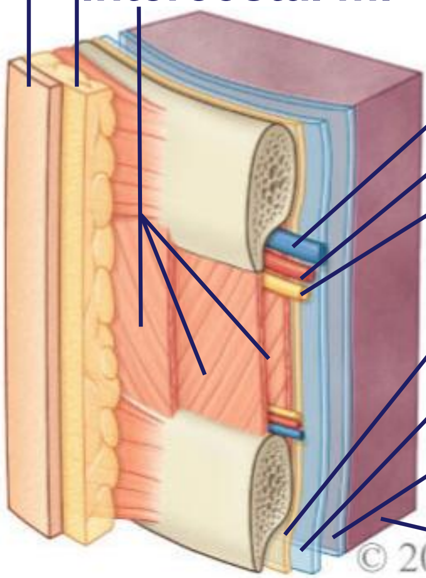


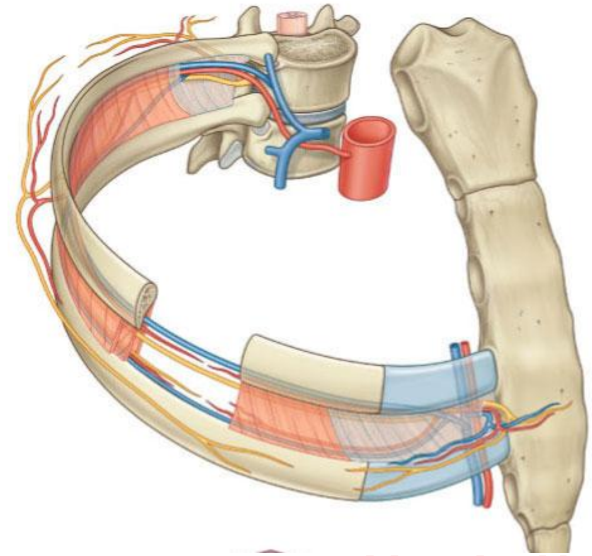
COSTOMEDIASTINAL

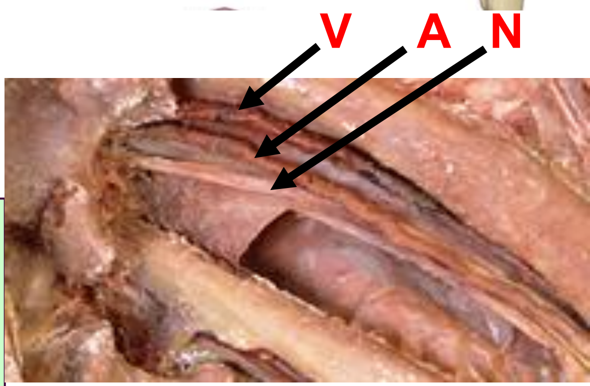


Neurovascular bundle and lymph

Neurovascular bundle and layers of the thoracic wall







Travels in costal groove on inferior aspect of rib

all incisions/tubes placed above rib to avoid VAN:

Thoracocentesis

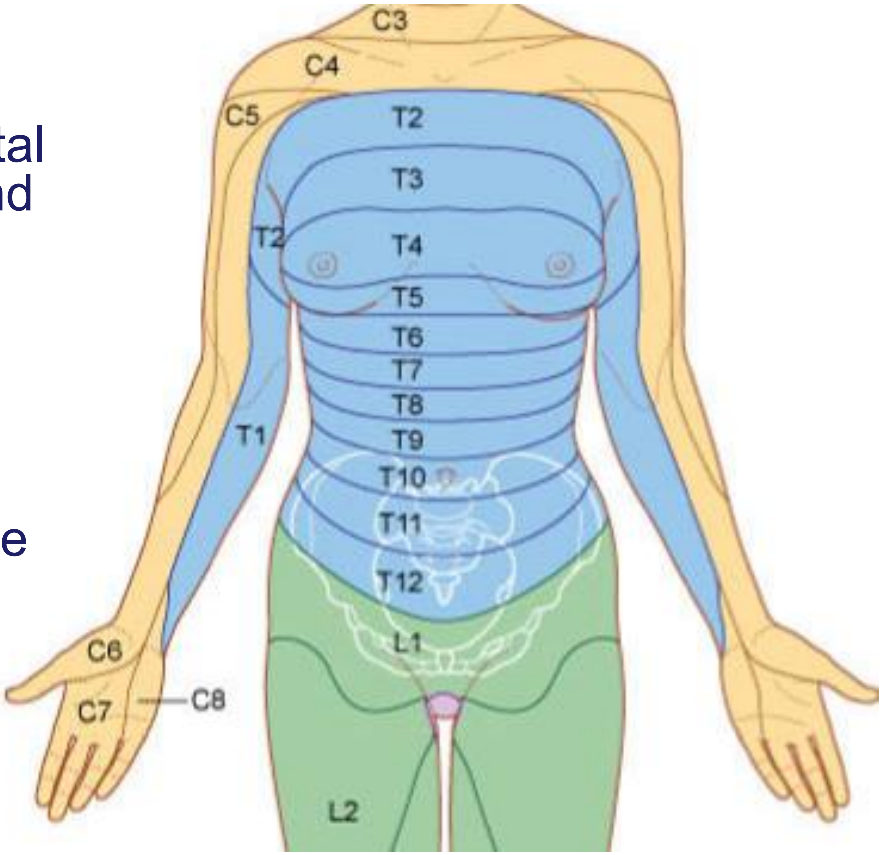
"pleural tap" to sample fluid from pleural space

Thoracostomy

Chest tube inserted to drain fluid or air from pleural cavity

Nerve supply to the thoracic wall

- Ventral rami of spinal nerves (intercostal nerves):
  - Motor innervation to muscles
  - Sensory innervation to ribs, costal cartilages, skin, costal pleura and peritoneum (T7-12)
- Dermatomes
  - Cutaneous branches from the spinal nerves supply the skin segmentally
  - Pain from the pleura is felt on the skin along the corresponding dermatome
- Some levels to remember:
  - T4 – Nipple
  - T7 – Tip of xiphoid
  - T10 – Umbilicus
  - T12 – Suprapubic region
  - L1 – Inguinal region



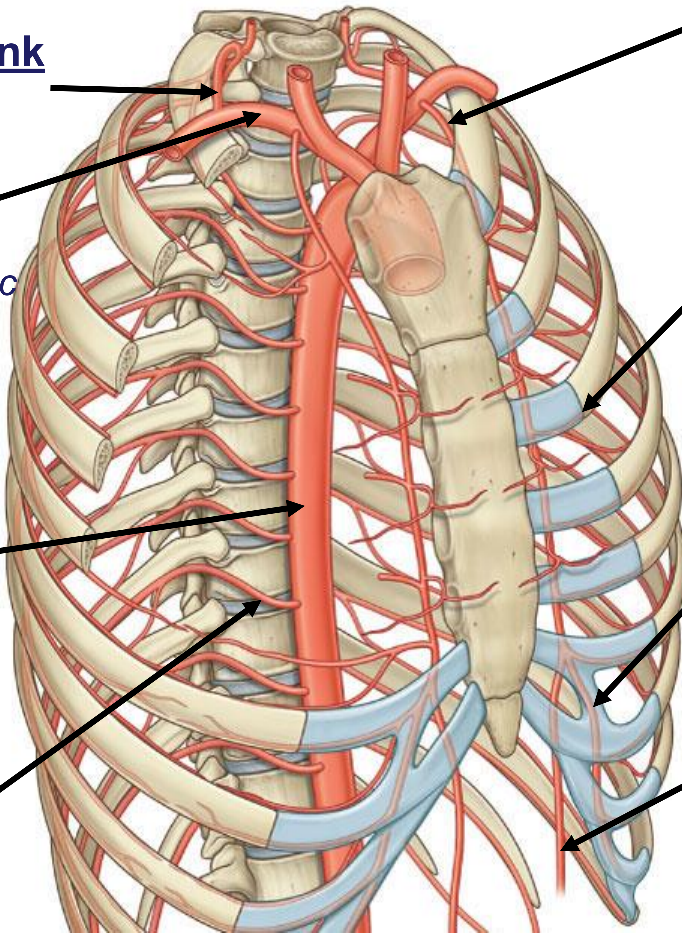
Arterial supply to the thoracic cage

Costocervical trunk
Supplies the first 2 posterior intercostal spaces

Subclavian A
Gives internal thoracic a.
Vertebral a.
Costocervical trunk

Descending Thoracic Aorta
Supplies posterior intercostal arteries

Posterior intercostal artery
Runs in costal groove



Internal thoracic A
Gives anterior intercostal arteries and terminates into musculophrenic and
superior epigastric

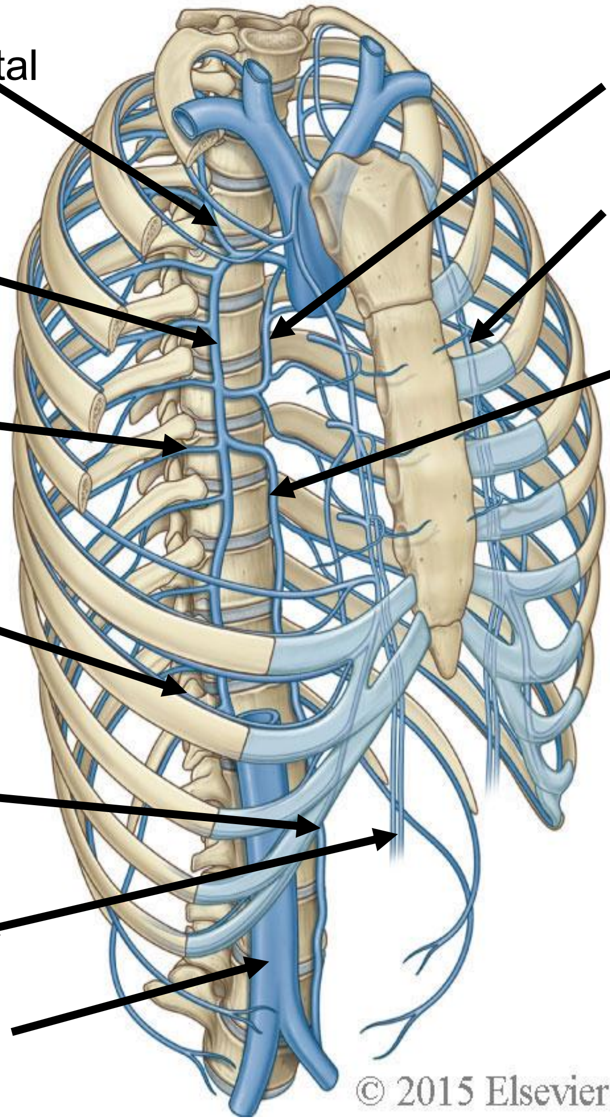
Anterior intercostal a.
Follows the inferior border of the rib and anastomoses with the posterior intercostal arteries

Musculophrenic a.
continues along the costal margin supplies diaphragm, anterior and lateral walls

Superior epigastric a.
continues onto the anterior abdominal wall and anastomoses with inferior epigastric a.

Parietal pleura
Supplied segmentally by the vessels associated with the adjacent wall
Diaphragmatic is supplied by superior phrenic arteries
Mediastinal portion is supplied by pericardiacophrenic artery

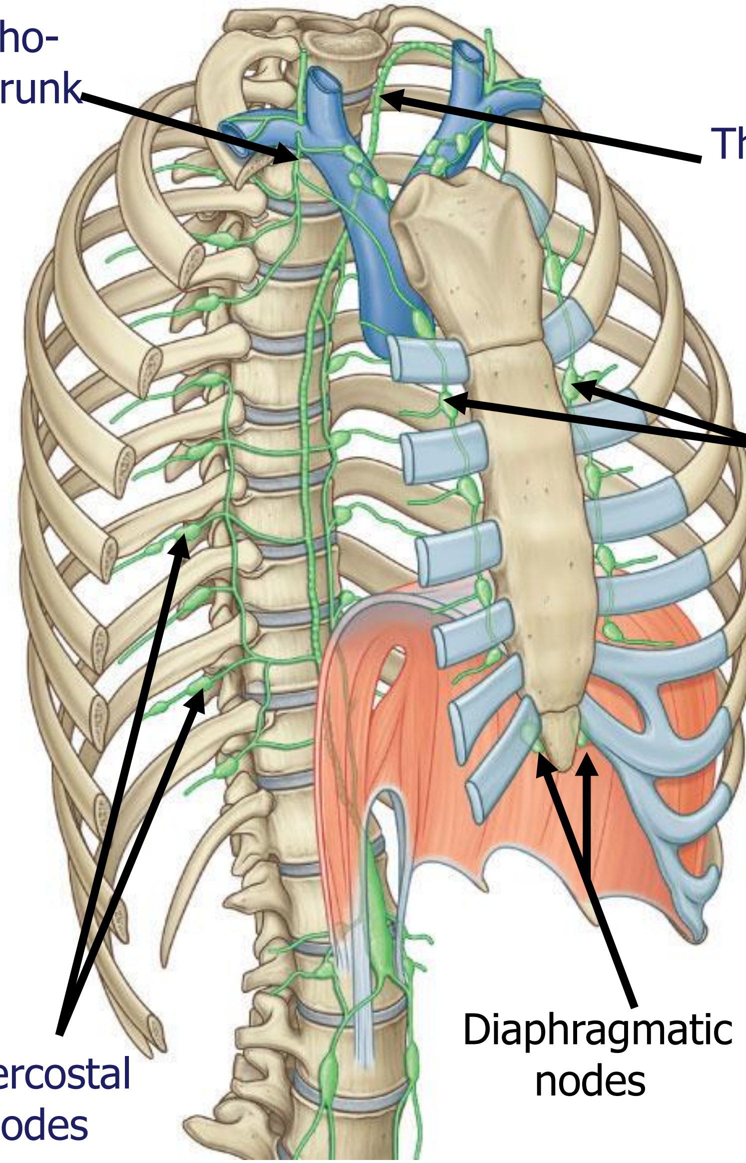
Venous drainage of thoracic wall



The veins from the upper 2 intercostal spaces posteriorly join together and may drain into the azygos and accessory hemiazygos or directly into the brachiocephalic vein

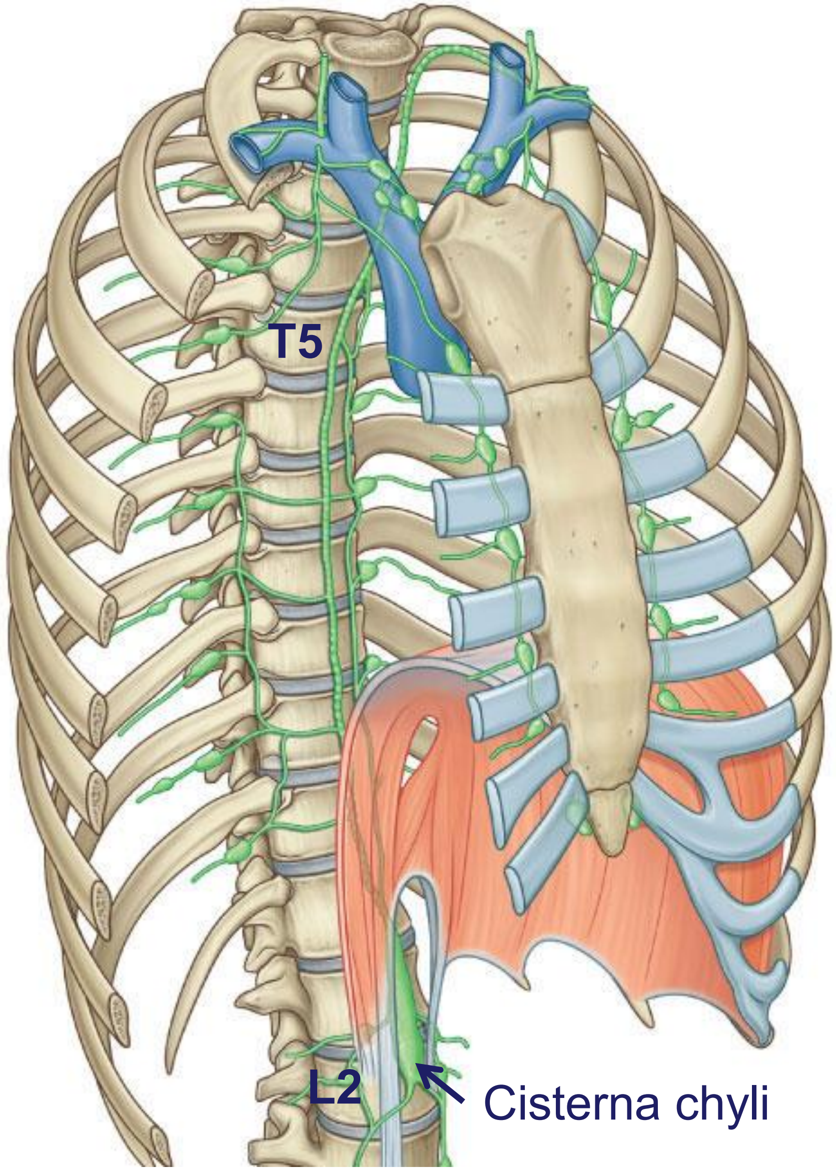
Lymphatic drainage of the thoracic wall

- Drainage:
  - Anteriorly to parasternal nodes
  - Posteriorly to intercostal nodes
  - Inferiorly to diaphragmatic nodes
- Most lymph nodes from thorax eventually drain into <u>thoracic duct</u>
  - Upper part of right thorax into right bronchomediastinal trunk



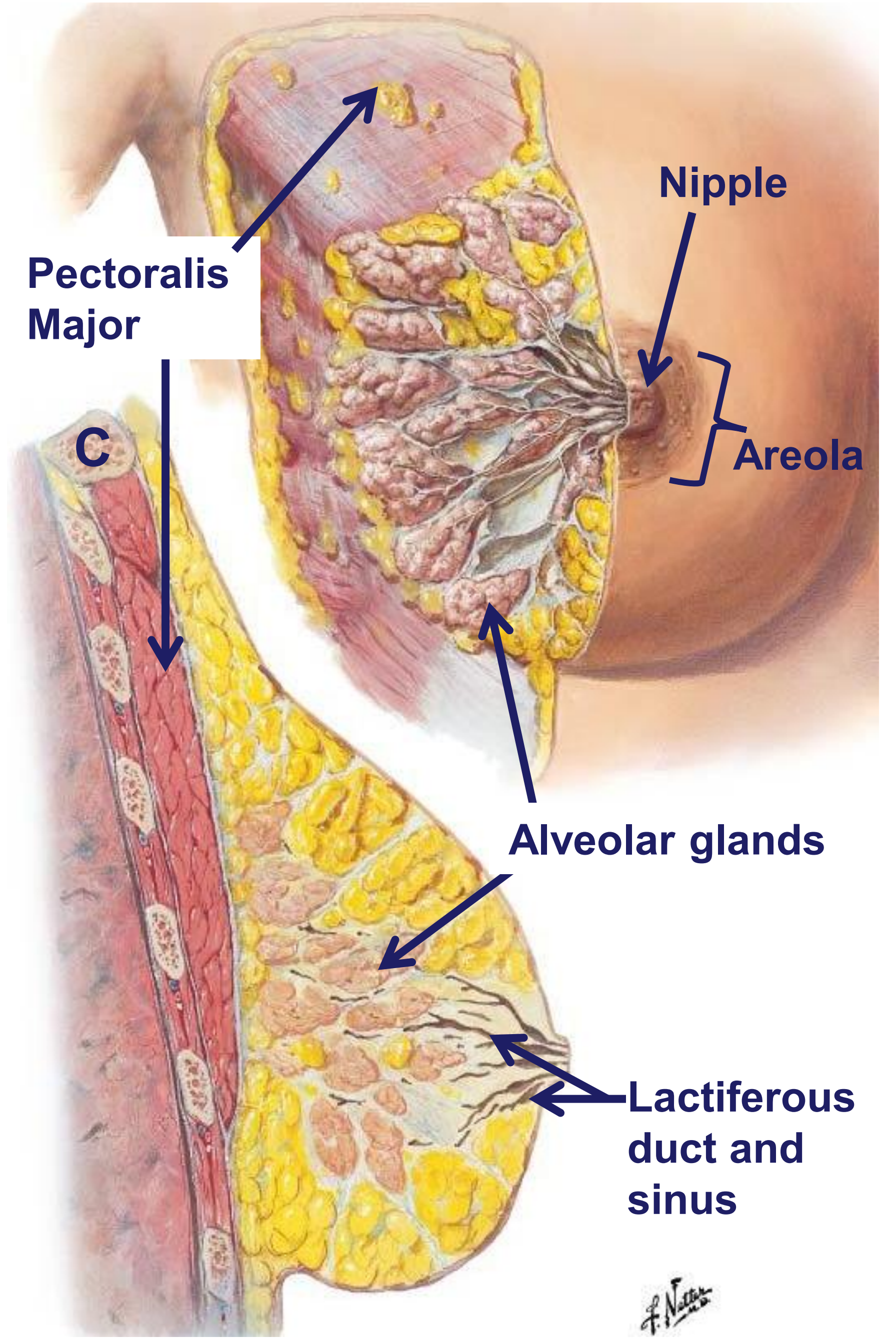
Thoracic duct

- The thoracic duct is the continuation of the cisterna chyli superior to the diaphragm
    - Thoracic duct pierces the diaphragm with the aorta
    - Runs between the aorta and azygos vein
    - Crosses to the left at T4/5
    - Empties into venous system at junction between IJV and SCV



Mammary gland

- Modified sweat glands that overlie the pectoralis major muscles
- Contains numerous connective tissue septa
    - This compartmentalizes the breast tissue
    - Some are condensed into ligaments = suspensory ligament (Suspends the breast)
- Numerous secretory lobules
    - The lobules increase in size during pregnancy and lactation
    - 15-20 lactiferous ducts open onto the nipple
    - Nipple is surrounded by the areola
- Surrounding the secretory lobules are areas of adipose tissue



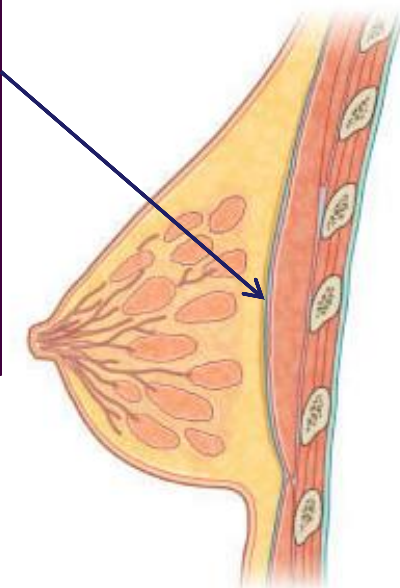
Mammary gland

Retromammary space

Space between the breast and the pec muscles

Allows for free movement of the breast

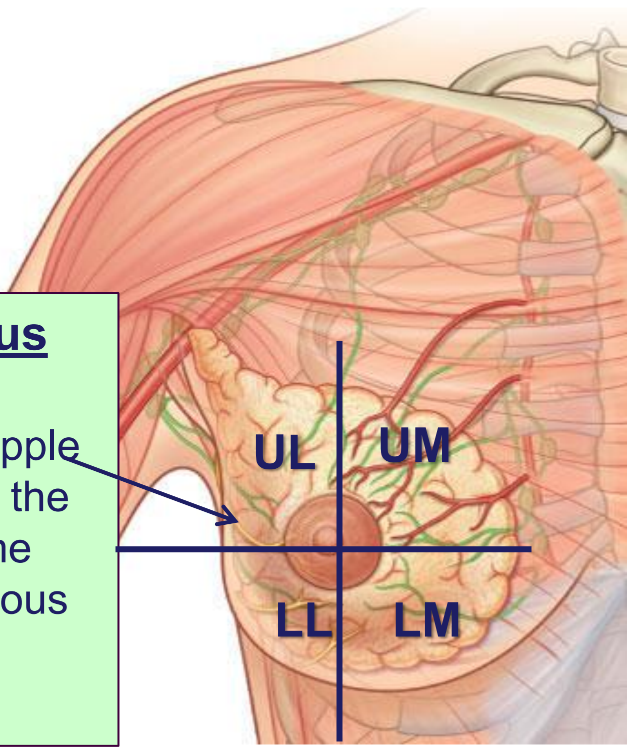
Breast is firmly attached to the pectoral fascia by the suspensory ligaments



T4 cutaneous nerve

Supply to the nipple corresponds to the T4 dermatome

Even in pendulous breasts



The breast is clinically divided into quadrants:

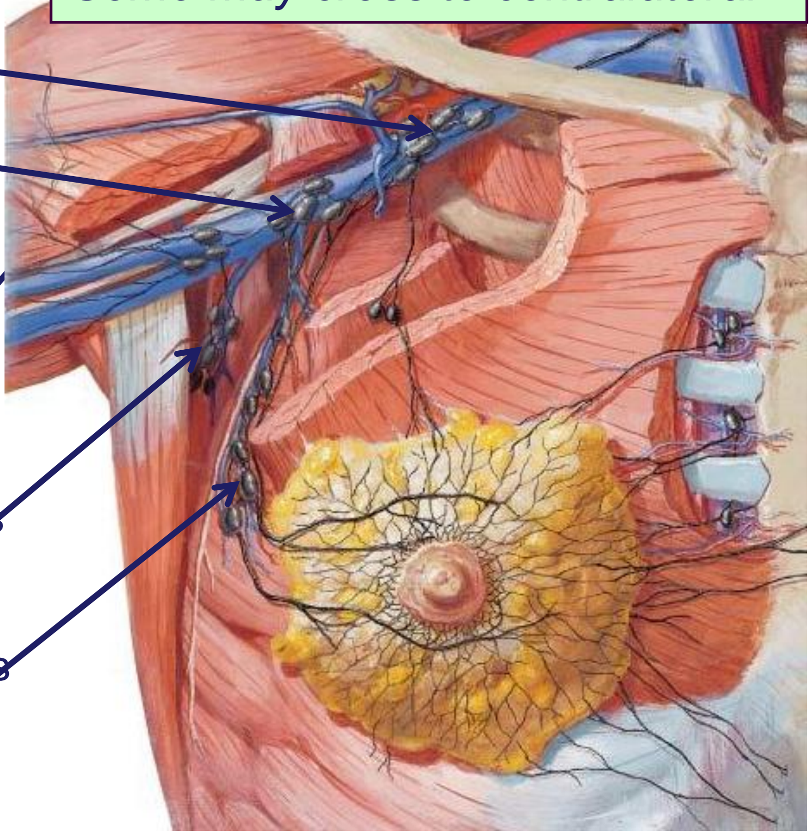
UL: Upper lateral
UM: Upper medial
LL: Lower lateral
LM: Lower medial

This is important as it relates to lymph drainage and therefore cancer metastases

Lymphatic drainage of the breast

Axillary nodes
Receives 75%

Parasternal nodes
Receives only 25%
Some may cross to contralateral



Lymph will drain **first to the nearest** group of nodes (typically **<u>anterior</u>**)

Continue to move from **node to node** until ultimately into the apical nodes

Eventually into bronchomediastinal and subclavian trunk

Joins the venous system by **left or right main duct**

<u>Sentinel node</u>
The first node into which lymph drains from any of the quadrants